CPPC 18/L

COMPUTER NETWORK & SECURITY

ACTIVITY No.9

SMTP & POP3

Submitted by:

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BS CpE 3B

Submitted to:

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Subject Professor

I. Objectives

- To configure email services using SMTP and POP3 on a server.
- To create email accounts for different client PCs.
- To understand how emails are sent and received within a network.
- To use DNS so that the mail server can be accessed using a domain name.
- To verify if the PCs can successfully send and receive messages.

II. Materials

- Cisco Packet Tracer
- 1 Server
- 1 Switch
- 2–3 PCs
- Laptop
- Copper Straight-Through cables

III. Procedures

1. Create the Network Topology

- Open Cisco Packet Tracer.
- Add the required devices in the workspace:
  - 1 Server
  - 1 Switch
  - 2–3 PCs
- Connect all devices using Copper Straight-Through cables.
- Connect the server to the switch.
- Connect each PC to the same switch.

- Make sure all connections are properly linked and active.

2. Configure the Server IP Address

- Click the Server.
- Go to Desktop.
- Open IP Configuration.
- Set the server with the following configuration:
  - IP Address: 192.168.1.1
  - Subnet Mask: 255.255.255.0
  - DNS Server: 192.168.1.1
- This allows the server to act as the mail server and DNS server in the network.

3. Enable the Email Services

- Click the Server.
- Go to the Services tab.
- Select EMAIL.
- Turn on the following services:
  - SMTP
  - POP3
- SMTP will be used for sending emails.
- POP3 will be used for receiving emails.

4. Set the Domain Name

- In the email service settings, enter the domain name:
  - Domain Name: email.com

- Click Set to apply the domain name.
- This domain will be used for the email addresses of the PCs.

5. Create Email Accounts

- Create the email accounts for each PC.
- Add the following usernames and passwords:
  - PC0_foilah
    - Username: foilah
    - Password: 123
  - PC1_krispatata
    - Username: krispatata
    - Password: 456
  - PC2_cess
    - Username: cess
    - Password: 789
- Click Add after creating each account.
- These accounts will be used by the PCs to send and receive emails.

6. Configure DNS

- Go to Server.
- Open the Services tab.
- Select DNS.
- Turn the DNS service ON.
- Add the following DNS record:
  - Name: email.com

- Address: 192.168.1.1

- Click Add to save the record.
- This allows the PCs to use test.com instead of typing the server IP address.

7. Configure Email on Each PC

- Click each PC one by one.
- Go to Desktop.
- Open Email.
- Enter the required mail settings for each PC.

For PC0_foilah

- Your Name: foilah
- Email Address: foilah@email.com
- Incoming Mail Server / POP3: mail.email.com
- Outgoing Mail Server / SMTP: mail.email.com
- Username: foilah
- Password: 123

For PC1_krispatata

- Your Name: krispatata
- Email Address: krispatata@email.com
- Incoming Mail Server / POP3: mail.email.com
- Outgoing Mail Server / SMTP: mail.email.com
- Username: krispatata
- Password: 456

For PC2_cess

- Your Name: cess

- Email Address: cess@email.com
- Incoming Mail Server / POP3: mail.email.com
- Outgoing Mail Server / SMTP: mail.email.com
- Username: cess
- Password: 789
- Click Save after entering the details.
- Make sure that the email settings match the account created on the server.

8. Send an Email

- Open the Email application on one PC.
- Click Compose.
- Enter the recipient email address, such as:
  - foilah@email.com
  - krispatata@email.com
  - cess@email.com
- Add a subject and message.
- Click Send.
- The email should be delivered to the selected recipient.

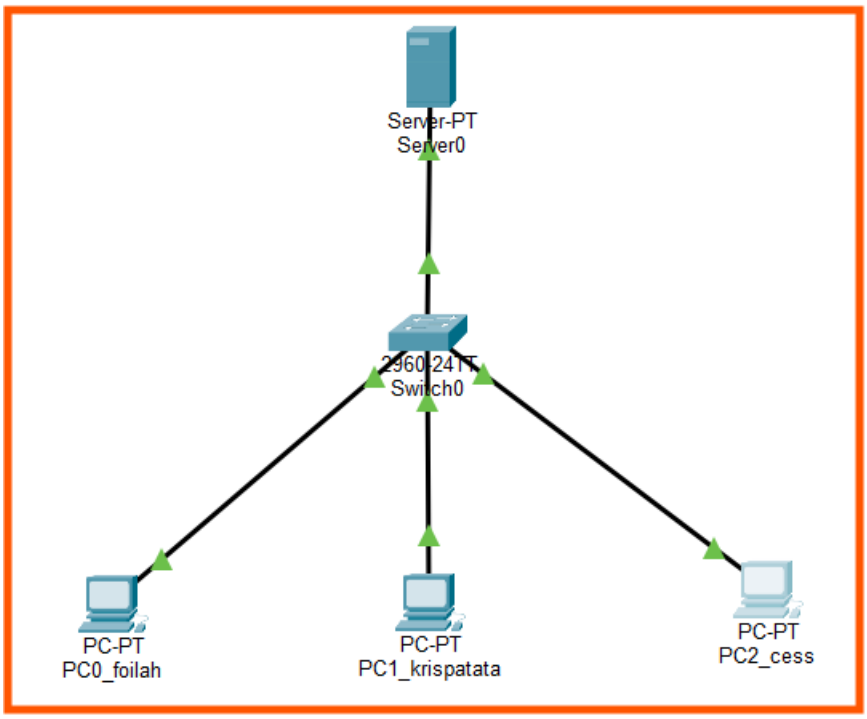
9. Receive an Email

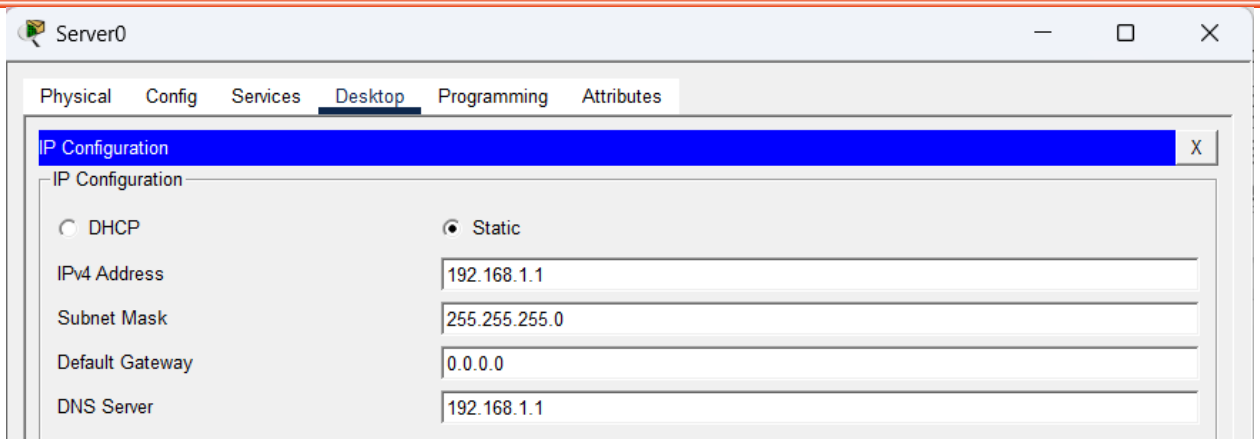
- Open the Email application on the receiving PC.
- Click Receive.
- Check if the message appears in the inbox.
- Open the received message to confirm that it was successfully delivered.
- Repeat the process between the other PCs to test email sending and receiving.

IV. Conclusion

The activity showed how SMTP and POP3 are used to send and receive emails within a network. By enabling the email services on the server, creating user accounts, and configuring DNS with the domain name "email.com", the PCs were able to communicate through email using their assigned email addresses. The successful sending and receiving of messages confirmed that the SMTP, POP3, and DNS configurations were properly set up and functioning correctly.

V. Documentation

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Physical
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Services
Desktop
Programming
Attributes
IP Configuration
X
IP Configuration
DHCP
Static
IPv4 Address
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Default Gateway
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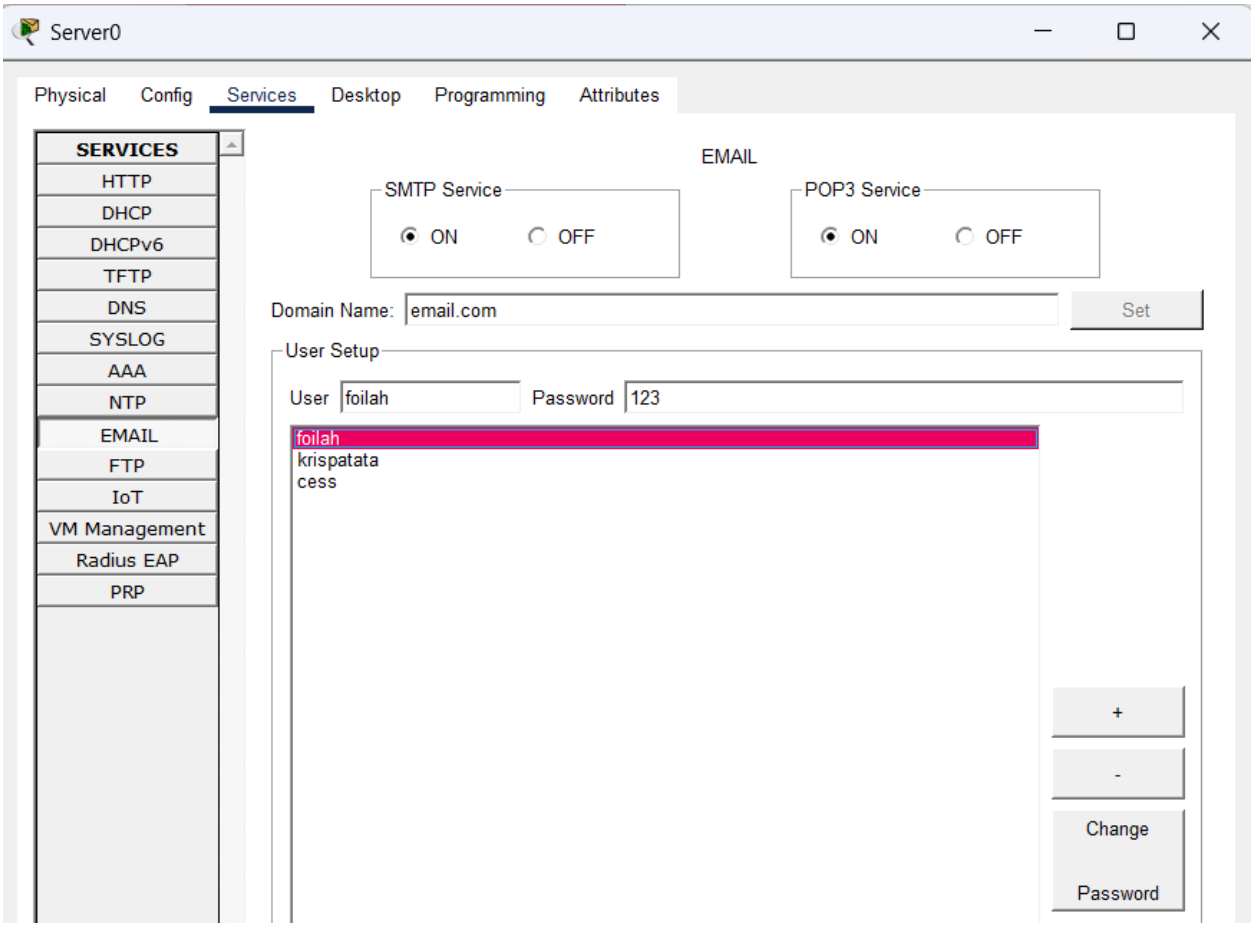
DNS Server
192.168.1.1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Services
Desktop
Programming
Attributes
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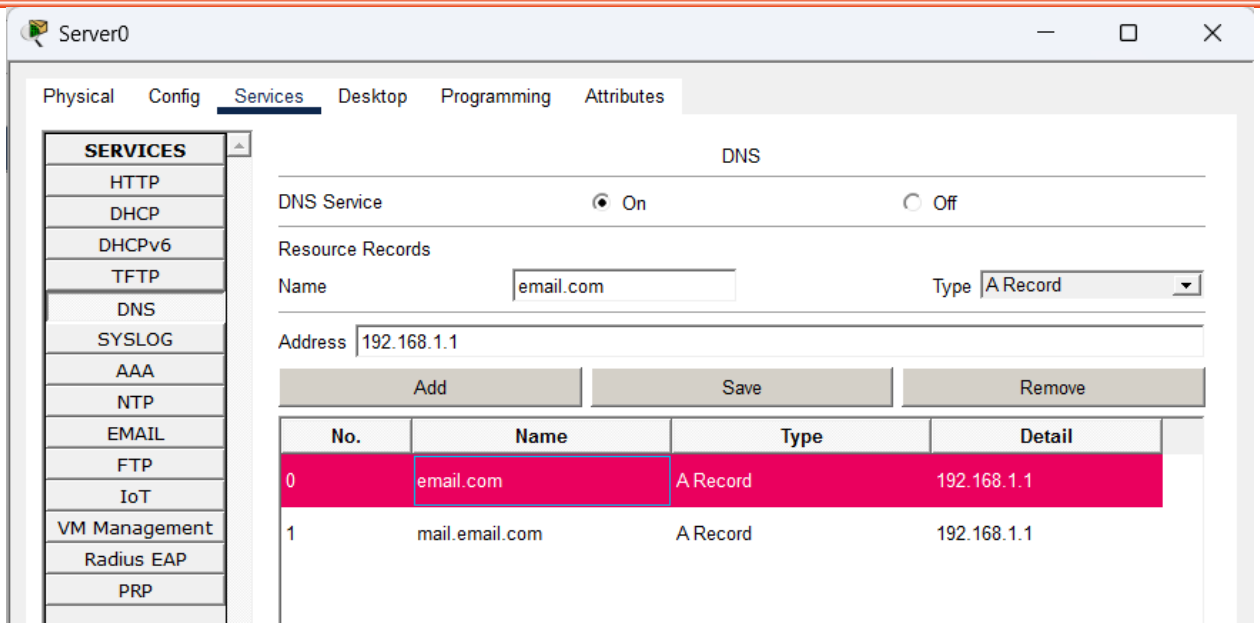
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
EMAIL
SMTP Service
ON
OFF
POP3 Service
ON
OFF
Domain Name:
email.com
Set
User Setup
User
foilah
Password
123
foilah
krispatata
cess
+
-
Change
Password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DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoT
VM Management
Radius EAP
PRP
DNS
DNS Service
On
Off
Resource Records
Name
email.com
Type
A Record
Address
192.168.1.1
Add
Save
Remove
No.
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Detail
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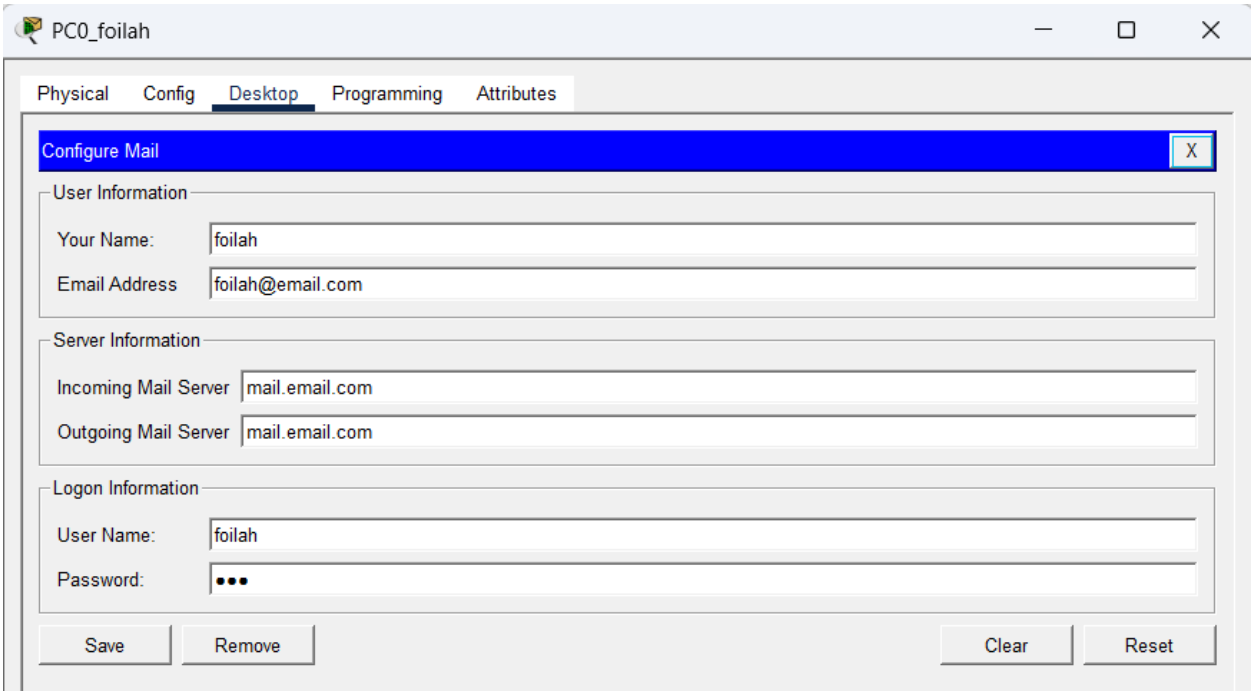
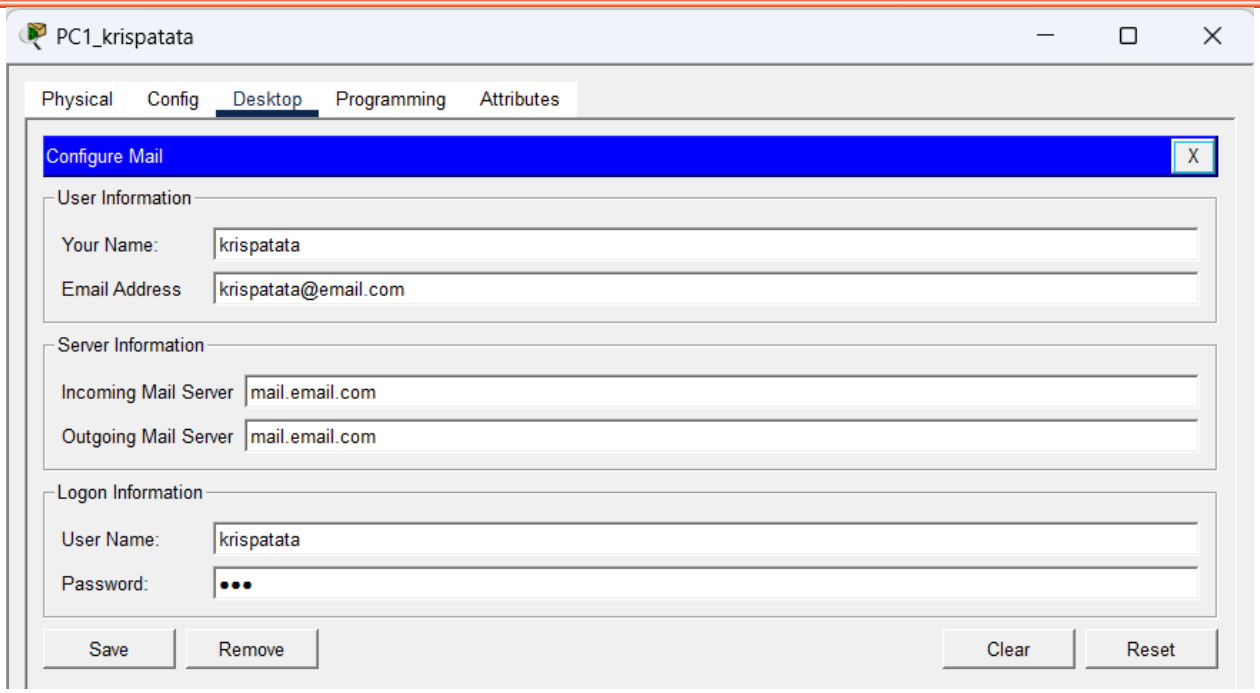
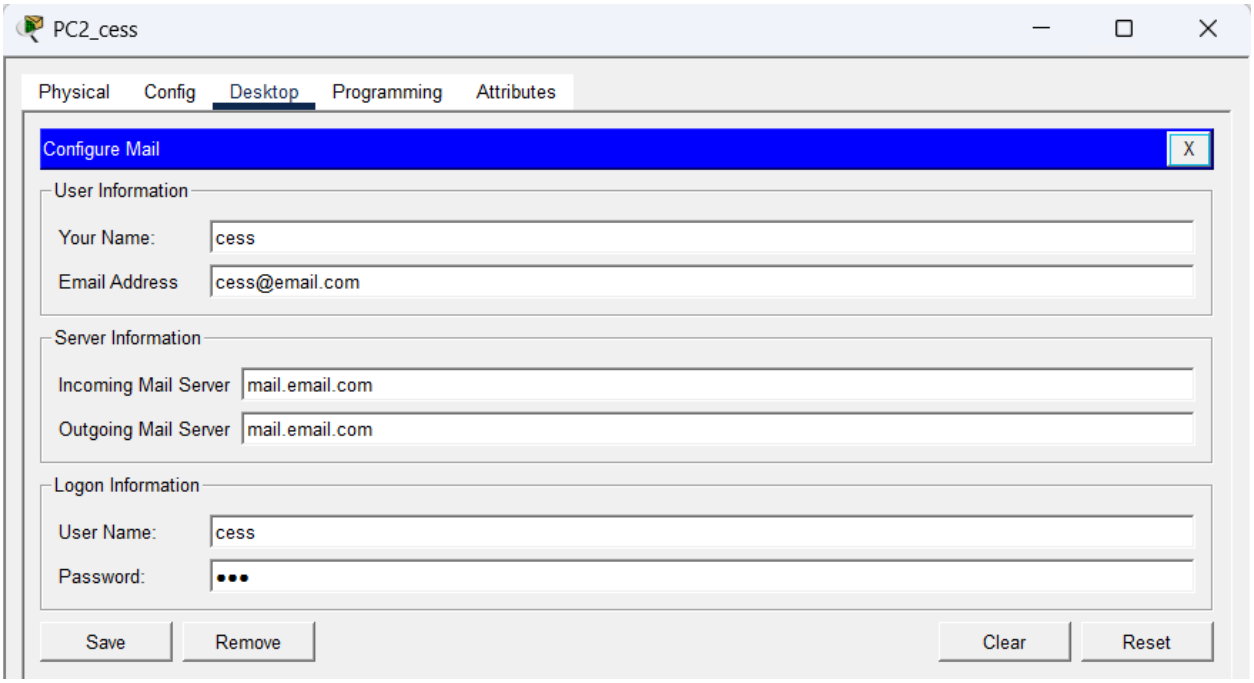
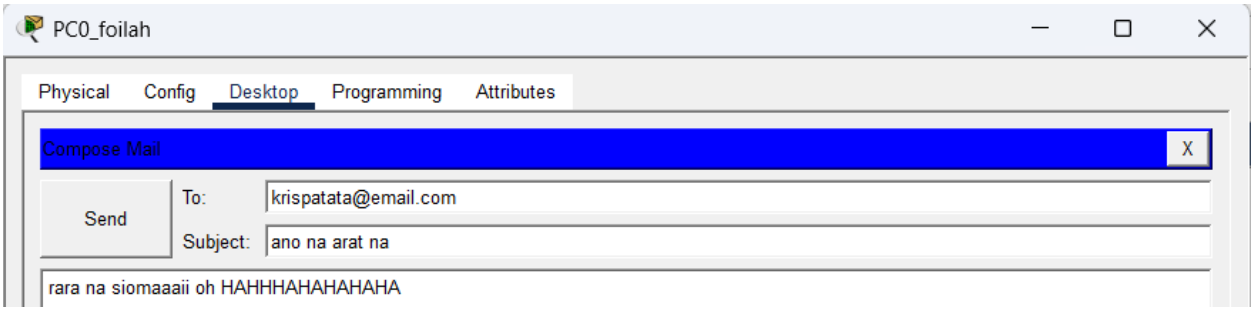

PC0_foilah
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name:
foilah
Email Address
foilah@email.com
Server Information
Incoming Mail Server
mail.email.com
Outgoing Mail Server
mail.email.com
Logon Information
User Name:
foilah
Password:
Save
Remove
Clear
Reset

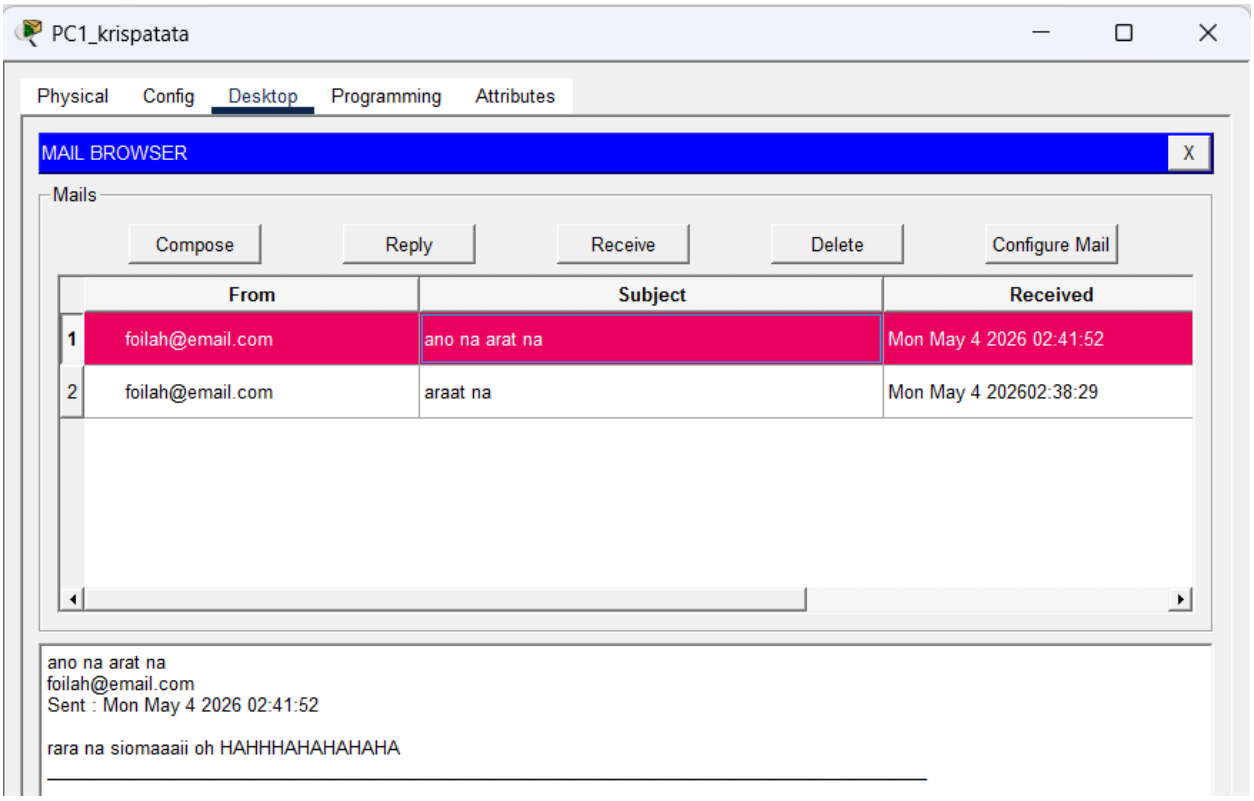

PC1_krispatata
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name:
krispatata
Email Address
krispatata@email.com
Server Information
Incoming Mail Server
mail.email.com
Outgoing Mail Server
mail.email.com
Logon Information
User Name:
krispatata
Password:
Save
Remove
Clear
Reset


PC2_cess
Physical
Config
Desktop
Programming
Attributes
Configure Mail
X
User Information
Your Name:
cess
Email Address
cess@email.com
Server Information
Incoming Mail Server
mail.email.com
Outgoing Mail Server
mail.email.com
Logon Information
User Name:
cess
Password:
Save
Remove
Clear
Reset

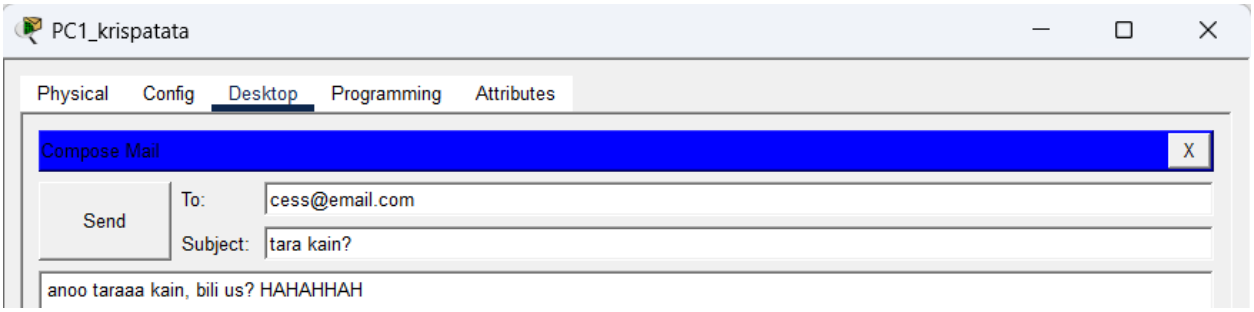
Composing email at PC0_foilah to be received by PC1_krispatata



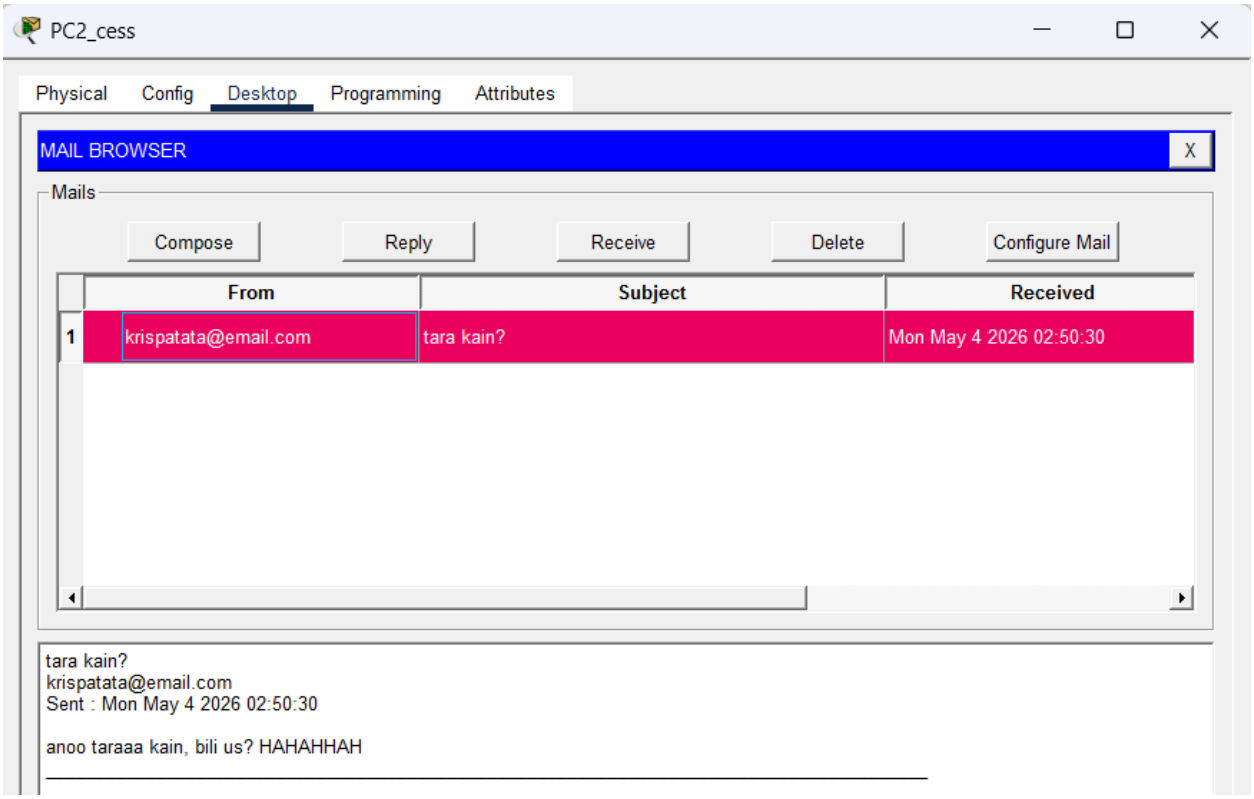
Receiving email at PC1_krispatata



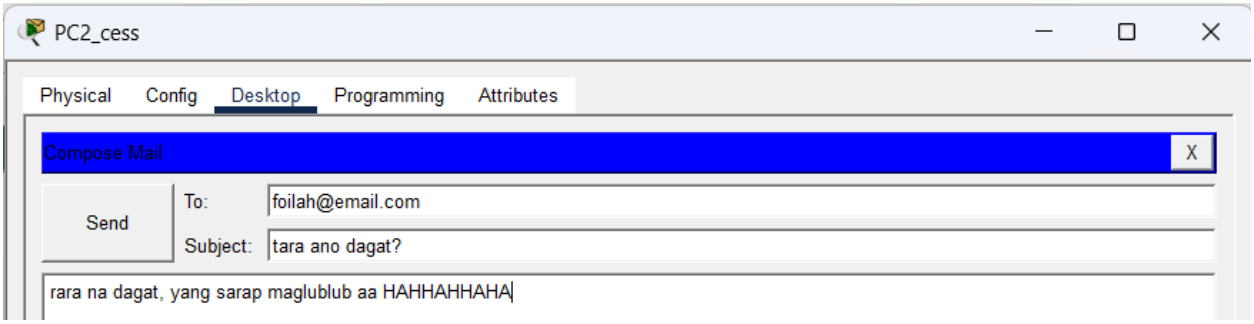
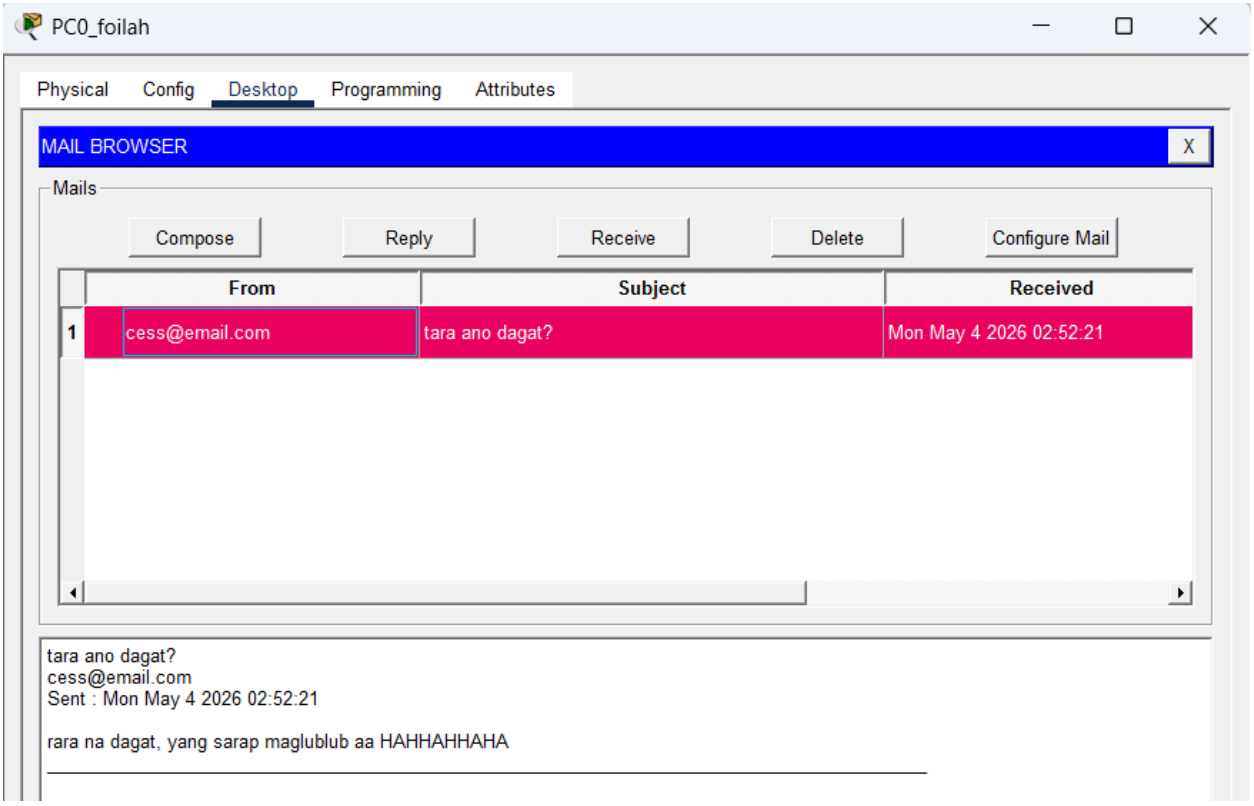
Composing email at PC1_krispatata to be received by PC2_cess



Receiving email at PC2_cess



Composing email at PC2_cess to be received by PC0_foilah



Receiving email at PC0_foilah